

AI Interview Orchestration Platform

Project Overview

The AI Interview Orchestration Platform is a fully automated recruitment and interview management system designed to reduce manual HR operations and streamline the complete interview lifecycle.

The platform enables candidates to register, apply for jobs, schedule interviews automatically, receive meeting links and notifications, and attend interviews with minimal human intervention.

The system also provides panel dashboards, interview history tracking, candidate status management, automated notifications, and workflow orchestration.

Project Goals

Primary Goals

- Automate interview scheduling
 - Reduce HR manual work
 - Generate Google Meet/Zoom links automatically
 - Notify candidates and interview panels automatically
 - Maintain interview history and status tracking
 - Support multi-round interviews
 - Build scalable workflow orchestration
 - Provide future AI integration capabilities
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Core Features

Candidate Module

Features

- Candidate registration
- Login and authentication
- Profile management
- Resume upload
- Document upload
- Job application
- View interview schedules
- Receive notifications

- View interview status
 - View interview history
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HR/Admin Module

Features

- Create job postings
 - Manage job requirements
 - View candidate applications
 - Assign interview panels
 - Configure interview rounds
 - Trigger interview workflows
 - Monitor interview progress
 - Manage candidate status
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Panel Module

Features

- View assigned interviews
 - Receive interview notifications
 - Access meeting links
 - View candidate profiles
 - Submit interview feedback
 - Update interview status
 - Track interview history
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Automation Features

Workflow Automation

The system uses workflow orchestration to automate:

- Interview scheduling
- Meeting generation
- Email notifications
- Push notifications
- Reminder scheduling
- Candidate status updates
- Multi-round workflows
- Retry handling

Technology Stack

Frontend

Technology	Purpose
Next.js	Frontend framework
React	UI development
Tailwind CSS	Styling

Backend

Technology	Purpose
FastAPI	API backend
Python	Core backend language
PostgreSQL	Database

Automation Layer

Technology	Purpose
n8n	Workflow orchestration
Redis	Queue management
Celery	Background workers

Notification Services

Service	Purpose
Resend/Brevo	Email notifications
Firebase Cloud Messaging	Push notifications

Meeting Integrations

Service	Purpose
Google Calendar API	Google Meet generation
Zoom API	Zoom meeting generation

System Architecture

```
graph TD; Frontend["Frontend (Next.js)"] --> Backend["FastAPI Backend"]; Backend --> Database["PostgreSQL Database"]; Database --> Automation["n8n Workflow Automation"]; Automation --> Services["External Services"]; Services --- GoogleMeet["Google Meet"]; Services --- Zoom["Zoom"]; Services --- EmailAPIs["Email APIs"]; Services --- Firebase["Firebase Notifications"];
```

Main Workflow

Interview Scheduling Flow

```
graph TD; A["Candidate Applies"] --> B["FastAPI Stores Application"]; B --> C["Workflow Triggered in n8n"]; C --> D["Panel Availability Checked"]; D --> E["Interview Slot Generated"]; E --> F["Google Meet/Zoom Link Created"];
```

```
graph TD; A[Email Notification Sent] --> B[Push Notification Sent]; B --> C[Interview Conducted]; C --> D[Panel Updates Feedback]; D --> E[Candidate Status Updated];
```

Queue System

Purpose of Queue

The queue system helps process background tasks asynchronously.

Tasks Handled by Queue

- Email sending
- Meeting generation
- Push notifications
- Reminder scheduling
- AI resume analysis
- Multi-user request handling

User Roles

Role	Responsibilities
Candidate	Apply jobs, attend interviews
HR/Admin	Manage jobs and workflows
Panel Member	Conduct interviews and submit feedback

Security Features

Planned Security

- JWT Authentication

- Role-based access control
 - OAuth integration
 - Secure file uploads
 - API validation
 - Encrypted credentials
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Future AI Features

Planned AI Integrations

LangChain

- Resume parsing
- Candidate scoring
- JD matching
- AI chatbot assistant

CrewAI

- AI recruiter agent
 - AI technical evaluator
 - AI HR evaluator
 - AI interview assistant
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Development Phases

Phase 1 — MVP

Features

- Candidate registration/login
 - Job application
 - Automated interview scheduling
 - Google Meet generation
 - Email notifications
 - Panel dashboard
 - Interview status updates
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Phase 2 — Scalable Automation

Features

- Redis queue
 - Celery workers
 - Push notifications
 - Reminder workflows
 - Multi-round interviews
 - Advanced scheduling
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Phase 3 — AI Features

Features

- Resume analysis
 - Candidate ranking
 - AI scoring
 - AI interview assistant
 - CrewAI multi-agent workflows
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Suggested Project Structure

```
project/
├── backend/
│   ├── app/
│   ├── routes/
│   ├── models/
│   ├── services/
│   └── requirements.txt
├── frontend/
├── automation/
├── database/
└── docs/
```

Deployment Plan

Local Development

- Windows
 - Node.js
 - Python
 - PostgreSQL
 - n8n
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Production Deployment

Recommended Environment

- Linux VPS
 - Docker
 - Nginx
 - PostgreSQL
 - Redis
 - SSL
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Conclusion

The AI Interview Orchestration Platform is designed to automate the complete interview lifecycle using workflow orchestration, backend APIs, notification systems, and future AI integrations.

The architecture is scalable, modular, and suitable for both MVP development and future enterprise-level expansion.